Gulab Kumawat

SOFTWARE DEVELOPMENT & AI SYSTEMS
+91 7051261687 | gulabgkg99@gmail.com | LinkedIn | GitHub | LeetCode

PROFESSIONAL SUMMARY

Information Technology undergraduate at NIT Srinagar with hands-on experience building full-stack and backend systems with GenAI/LLM integration in JavaScript and Python, and a strong DSA foundation in C++; seeking software engineering roles.

EDUCATION

National Institute of Technology, Srinagar 2023 - 2027
B.Tech in Information Technology | CGPA: 8.2/10

Rajasthan Board of Secondary Education 2021 - 2022
Senior School Examination (Class XII) | Percentage: 94.20%

PROJECTS

AI Database Agent (Text-to-SQL) **Source code** June 2026
FastAPI, LangGraph, PostgreSQL, SQLAlchemy, React, Docker, AWS

- Built a full-stack text-to-SQL agent that compiles natural-language questions into safe, read-only queries through a multi-node LangGraph pipeline.
- Enforced query governance with an sqlglot AST validator: SELECT-only, auto-injected row limits, and statement timeouts. Blocked unsafe SQL before database execution; secured access with JWT, RBAC, and Fernet-encrypted credentials at rest.
- Containerized FastAPI, React, PostgreSQL, Redis, and Nginx with Docker Compose and deployed to AWS EC2 behind a reverse-proxy gateway, using an async SQLAlchemy backend and Alembic migrations.

Accident Risk Prediction August 2026
Python, Pandas, Scikit-learn, XGBoost

- Developed a machine learning system to predict road accident risk from speed, weather, traffic, road condition, time, and historical accident data, using preprocessing and engineered rush-hour and weather-severity features.
- Trained and compared Logistic Regression, Random Forest, and XGBoost classifiers. Built a dashboard for real-time Low, Medium, and High risk predictions from user-entered road conditions.

eMitra AI - Multi-Source RAG Engine **Source code** April 2026
FastAPI, React, Gemini, Qdrant, Crawl4AI

- Built a multi-source Retrieval-Augmented Generation (RAG) engine that answers queries using uploaded documents, crawled website content, and live web search, with semantic retrieval from Qdrant.
- Developed an end-to-end ingestion pipeline for website crawling, document parsing, semantic chunking, embedding generation, and vector indexing, enabling low-latency, citation-based responses while reducing LLM hallucinations.

TECHNICAL SKILLS

Languages: C++, JavaScript, Python, SQL
Databases: MongoDB, PostgreSQL, Redis
AI / ML: Machine Learning, Deep Learning, LangChain, LangGraph, Gemini API, RAG
Libraries & Frameworks: React.js, Node.js, REST APIs, Express.js, FastAPI
Tools & Cloud: Git, GitHub, AWS, Docker
Core Subjects: DBMS, OOP, Data Structures & Algorithms, Machine Learning, Operating Systems, Computer Networks

CERTIFICATIONS & ACHIEVEMENTS

- Completed Microsoft Generative AI Course. View certificate
- Solved **800+ DSA problems** across LeetCode and GeeksforGeeks.
- **TechM Code 2026 Finalist** - national-level hackathon by Tech Mahindra.